Simulation and Control Laws of the Flying Model Unifier-19

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***Abstract*—The number of emerging projects with Distributed Electric Propulsion (DEP) has significantly increased these last five years. This is undoubtedly, driven by the strong ambition to decarbonize the aviation industry and the needs to flight more passenger while reducing energy consumption. Only, there is not lots of literature on the flight control and handling qualities of these DEP aircraft. In respect to this, this paper aims at contributing on Control Augmentation System that may apply to DEP aircraft. Thus, a pitch-rate CAS loop has been proposed to study the handling qualities of DEP aircrafts in the longitudinal axis. The proposed Pitch-rate CAS loop is verified by linear flight simulations, where all dynamics of the UNIFIER19 aircraft are considered.**

***Index Terms*—Distributed Electric Propulsion, Pitch-rate, Control Augmentation System, Longitudinal control, Feedback loop**

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I. INTRODUCTION

Over the past decade, the aviation industry has come under increasing pressure from the public and institutions to reduce the carbon footprint of air transport industry. Indeed, in 2022, a study 1 by IEA, reveal that 2% of the global Carbone emission in the atmosphere are from the aviation industry. As answer, aircraft manufacturers and research instutions like NASA are investigating new concept that could help reduce footprint of air transport industry.

Several ongoing projects aims at improving aircraft performance, as well as fuel consumption and noise reduction. Distributed Electric Propulsion is one of the key concepts proposed by a research program of NASA called N3-X featuring advanced Electrified Aircraft Propulsion technologies. [1]

The N3-X aims both at reducing fuel consumption, thanks to electric propulsors, and increasing aircraft aerodynamic performance, thanks in part to more aerodynamic geometries. The N3-X combines two innovative concepts: the use of electric thrusters and the distribution of engines on the wing.

The use of electric motors for propulsion is not new. Throughout history, we've seen prototype gliders using electric thrusters. In fact, advances in the production and storage of electricity on board aircraft have paved the way for the growing use of control systems and even electric propulsion. Electric propulsion alone, or as a complement to conventional fuel propulsion, reduces the amount of energy consumed per passenger carried. What's more, as battery storage capacities continue to grow, they can also be used to extend aircraft range. [2]

The concept of distributed propulsion, on the other hand, is relatively recent, and it is difficult to date it precisely, as its development has been gradual. Unlike traditional propulsion

systems, where the motors and therefore the thrust are grouped together at a single point, distributed propulsion involves distributing the propulsors along the leading edge of the wing. This approach provides better thrust distribution and offers significant advantages in terms of control, efficiency and safety. [2] Yet, no aircraft with this type of propulsion is currently on the market, since the introduction of this propulsion requires the development of suitable control laws providing effective operating conditions, especially in terms of directional stability and control. [2]

The FAA's Aviation Maintenance Technician Airframe Handbook defines stability as: "The characteristic of an aircraft that causes it to return to its original flight condition after it has been disturbed". When it comes to aircraft stability study, the subject is often divided into two subjects: static and dynamic stability.

The American Institute for Astronautics and Aeronautics defines Static stability as "the initial tendency of an airplane, following a perturbation from a steady-state flight condition, to develop aerodynamic forces or moments that are in a direction to return the aircraft to the steady-state flight condition".

In other hand, dynamic stability is defined as the response of the aircraft to disturbance over time. Each of these categories can have a major impact on aircraft longitudinal directional characteristics. Most of the modern aircrafts incorporates feedback control systems that augment the dynamic stability of the aircraft. The feedback control systems are developed as Inner loop grouped into three broad categories: stability augmentation systems, control augmentation systems and fly-by-wire systems.

This research work is part of a study conducted at SUPAERO to develop control laws for Distributed Electric Propulsion. Initial work on Lateral Stability Control was carried out by Jeanne Trias on the UNIFIER19 model. In this paper, we will continue the study Control Augmentation Systems, focusing on directional-longitudinal control laws.

In this paper, we will first review the literature to understand concepts and the advantages behind Distributed Electric Propulsion (DEP), then we will present the classical pitch-rate Control Augmentation System (CAS) to study its structure. Based on that study, a pitch-rate CAS for DEP aircraft will be proposed.

II. Distributed Electric Propulsion Concept

As said in the introduction, DEP combines two the concepts: Distributed Propulsion (DP) and Electrical Propulsion Aircraft (EPA). The Distributed Electric Propulsion concept represents a paradigm shift in aircraft propulsion technology.

A. Introduction to Concepts behind the DEP: Distributed Propulsion and Electric Propulsion Aircraft

1) Distributed Propulsion:

: Instead of relying on one or two jet engines mounted on the wing or attached to the fuselage, the DP concept relies on many smaller propulsor mounted along the airframe. There is not yet a standard definition of Distributed Propulsion but in this paper, we will consider a Distributed Propulsion as the repartition of Propulsion Units (motors or fan) along the airframe in the aim of improving overall aircraft efficiency and capabilities by improving aerodynamic performances of the air vehicle. [1]



Fig. 1: Artist rendering of the N3-X concept aircraft with hybrid wing body airframe. Credits: NASA

2) Electric Propulsion Aircraft concepts [1]:

: In other to reduce fuel and energy usage in aviation, the NASA Glenn's research centre is developing three (3) innovative aircraft propulsion concepts, including all-electric aircraft, Hybrid electric aircraft and Turboelectric aircraft.

- All-Electric: The All-Electric aircraft concept relies the electricity provided by on onboarded batteries as sole source of propulsion power.

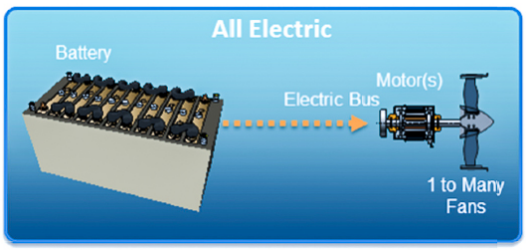


Fig. 2: All-Electric

- Hybrid:In this configuration the aircraft is powered by several energy sources used, either alternately or in combination.During the flight, the onboaded batteries can either supply the actuator or be charged by the charge the jet engine turbine. Three main configurations are under study:
  - Parallel hybrid : In this architecture, both the battery-powered motor and the turbine engine are mounted on a shaft that drives a fan. In that case, they can work alternately or in tandem to provide propulsive power.

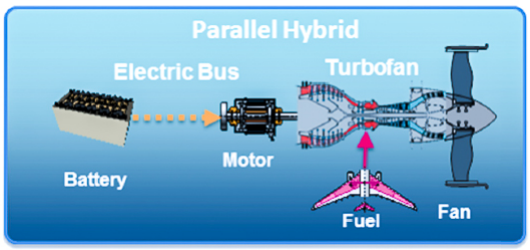


Fig. 3: Parallel hybrid

  - Series hybrid : In the series hybrid configuration, the battery-powered motors are the only one connected to the fans. The jet fuel turbine is then used to charge or to drive batteries by using and electrical generator.

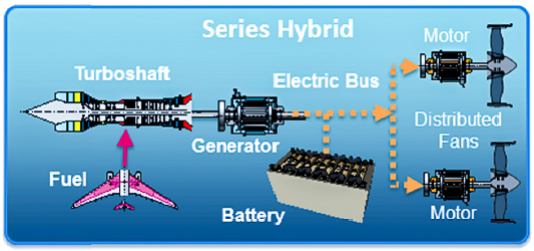


Fig. 4: Series hybrid

  - Series/parallel partial hybrid: This architecture uses one or more fands that can be driven directly by a jet fuel engine, and additional fans that rae driven exclusively by electric motors powered by batteries or a turbine-driven generator.

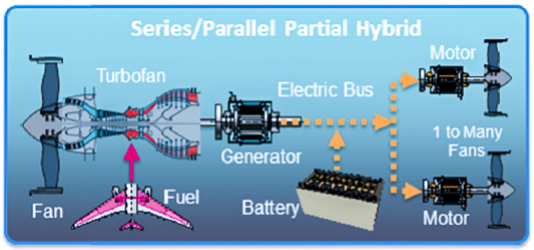


Fig. 5: Series/ parallel partial hybrid

    The combination of jet fuel or SAF with electricity provided by batteries or fuel cells- enables the aircraft to optime energy consumption and increase overall efficiency. Studies conducted by Airbus on hybrid aircraft have shown that hybridisation could help reducing the fuel consumption by up to 5% compared to traditional flight.
- Turboelectric architecture: [NASA] In this architecture the propulsive power does not come from batteries. Instead of using batteries as propulsive-power source in flight, turboelectric aircrafts use gas turbines to power electric generators, which in turn power inverters and individual direct current (DC) motors that power separate distributed electric fans.
  Two configurations of turboelectric aircraft exits :
  - Partially turboelectric: In this architecture, only part of the propulsive power is provided by the electrical propulsion system, the rest is provided by a turbofan driven a by gas engine.

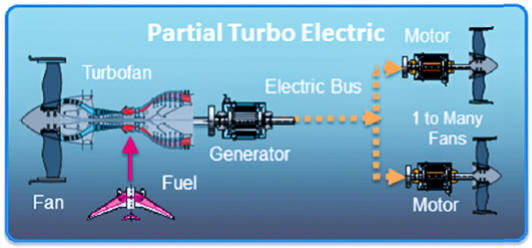


Fig. 6: Partially turboelectric

  - Fully turbo electric: Here, the gas turbine engine is used to drive electric generators witch then power inverters and motors to drive disturbed fans. The

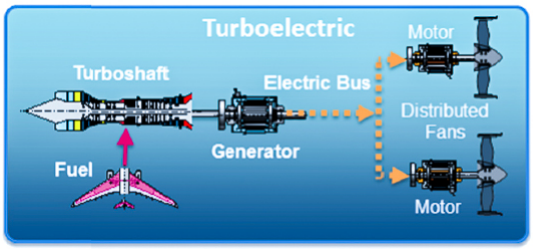


Fig. 7: Fully turboelectric

    DEP architecture is thus the using electric power to drive multiple motor-driven propulsors fans.

B. Advantages of Distributed Electric Propulsion

Improving the overall aircraft performance is not the only benefit we could get from the DEP aircraft. The increasing number of propulsion unit (propeller or fans) near the aircraft's aerodynamic surfaces create aero-propulsive coupling due their tight integration. [Review] These coupling could be of a great advantage if a DP system is carefully.

Inherent redundancy: the distribution of many propulsors across the aircraft reduces the size of the system by minimizing the need for redundancy. Thus, the loss of thrust due one engine failure can be compensated by increasing the thrust generated by the remaining propulsors. [8]

Optimal allocation of propellers: strategic placement of propulsors can increase the controllability of the aircraft by using differential thrust. Indeed, the tight distribution of the propulsion across the aircraft can allow the Boundary Layer Ingestion (BLI), thus reducing energy consumption. Also, differential thrust allocation can be used to increase the pilot control authority on the aircraft longitudinal and lateral axis and reduce the use of control surfaces like the elevator and the rudder. Reducing control surfaces can help decrease the aircraft global weight. [8]

Higher bypass ratio: DEP aircraft have a higher Bypass Ratio (BPR) due the large number of propulsors on a DEP aircraft. Indeed, as the DEP concept uses larger number of propulsors, the total air mass flow through the propulsors is also augmented. This considerably increases efficiency and makes the flight quieter. [2]

No power lapse at altitude for all-electric configuration: Electrically propelled motor aircrafts do not experience power loss altitude, unlike traditional jet engine because of decreased air density. Therefore, the DEP's available power would not be affected by changes in altitude or temperature with the proper thermal management. [8]

Shorter take-off and landing: by using the propulsor's slip-stream interaction with the aerodynamic surfaces, the lift or control authority on the aircraft is increased. The supplementary produced lift enhances vehicle low speed performance and can potentially reduce the take-off distance by up to 50%. By diverting the exhaust flow downward at the trailing edge, DEP aircraft can utilize the Coanda effect to convert a portion of the thrust into lift. [9]

To summarize, the DEP aircraft have substantial impact on the overall aviation industry.

III. Control Augmentation Systems

Distributed propulsion increases the aircraft performance as it increases the pilot control authority thanks to the fan or propeller slipstream interactions with aerodynamic surfaces to produce enhanced lift. [Review] The level of control authority depends on the type of propulsors (e.g.: propeller or ducted fans) used, their pitch angles, their distance between them. This introduces, the concept of Propulsion-Controlled-Aircraft (PCA). DEP can also be used as a method to control the vehicle's attitude thanks to the creation of a thrust differential across the wing.

A. Introduction to flight control [6]

From the Wright brothers to the Airbus A350 the flight control and handling qualities have always been a core aspect of the commercial aircraft design process. This has been done through a systematic design process and the emphasis placed on making the aircraft controllable by the pilot rather than inherently stable. [LIVRE] While the first aircrafts were controlled by classical control – consisting of "successive loop closure guided by a good deal of intuition and experience- modern aircraft, have three levels of automatous flight control:

- Autopilot: this is meant to perform specific tasks to help the pilots during long distance flight or during the critical stages of the flight. It is designed to satisfy requirements on steady-state error and disturbances rejections. For instance, to help the pilot hold an altitude and air-speed flight for a long-distance flight, the Altitude Hold/Mach Hold feed-back loop have been developed.
- Stability Augmentation Systems: it provides artificial damping to improves the responsiveness of the aircraft to manoeuvrings commands. Basically, it is an electromechanical device that senses undesirable motions like -wind shear, gusts- and powers the appropriate control surfaces to damp out the motion. [Flight Stability] Conventionally, stability augmentation laws are designed separately for longitudinal dynamics and lateral-directional dynamics by decoupling the aircraft dynamics in considered flight conditions. For example, the pitch-axis stability augmentation provides a satisfactory natural frequency and damping for short-period mode. The variables alpha and pitch-rate are used to modify the frequency and damping.

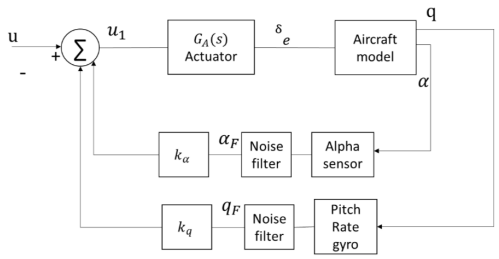


Fig. 8: Pitch Stability Augmentation

- Control Augmentation Systems: as it intends to make the aircraft returns to its equilibrium position, the SAS reduces the control authority of the pilot. In case one wants, high manoeuvrability and agility, specialized CAS must be developed. With CAS, the aircraft dynamic response is typically well-damped, and control response is scheduled with the control systems gains to maintain desirable characteristics throughout the flight envelope.

B. Pitch-Rate Control Augmentation System

To increase their manoeuvrability and agility, modern aircrafts use specialized CAS in addition to traditional SAS. The Pitch-rate CAS intends to enhance damping and generating synthetic data for identification based on a typical configuration of feedback control. CAS are then designed to enable situation that requires precise tracking and critical flight phases such as approach and landing.
The next figure is a block diagram of pitch-rate CAS. The general structure of conventional pitch-rate is represented.

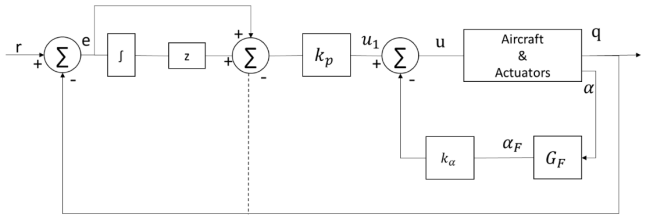


Fig. 9: Pitch Rate Control Augmentation

The block diagram consists of two nested feedback loops. In the inner loop, the AoA signal is fed back, with two static controller gain, kα and Gf in the feedback path. In the outer loop, the pitch-rate error is fed back through a proportional-integral (PI) controller, with transfer function.

$$C_o(s) = K_p + \frac{K_i}{s} \tag{1}$$

Therefore, proportional-plus-integral compensation is used to provide more precise control. The proportional path of the PI compensator can be replaced by an equivalent inner-loop pitch-rate feedback shown as a dotted line.

In this control system, the inner loop provides the dynamic stability for longitudinal motion while the outer loop enables the control of pitch-rate.

IV. Simulink Model and Controls Laws Implementation

A. UNIFIER19 aircraft model [5]

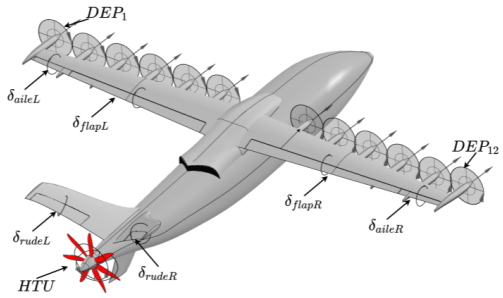


Fig. 10: Effectors of the UNIFIER19 C7A-HARW

UNIFIER19 is a partnership project that aims to assess a hypothetical 19-seat miniliner powered by hydrogen. As part of the project flight control laws and a flight dynamics simulator were created.
The UNIFIER19 C7AHARW effectors are shown in Figure 8. The vehicle has six separate control surfaces, as can be seen. On the leading edge of the vehicle are 12 fixed pitch independent Distributed Electric Propellers (DEP). Finally, the vehicle contains one variable pitch main propulsors, also called a horizontal thrust unit (HTU), mounted at the aft of the fuselage.
Finally, UNIFIER19 is equipped with a V-tail, therefore, yaw and pitch movements are correlated. Because it is not feasible for a pilot to oversee every effector on a vehicle like the C7A-HARW, an advanced fly-by-wire system is used.
All these factors are modeled in the model below in the subsystem C7H_HARW [2] from GitHub. We are not getting into details for this subsystem. In fact, the only subsystem we are interested in is the FCC subsystem, which is where we will implement our control laws. We only need the sensors bus to create those laws; we can also use the sidestick input, but this is optional.

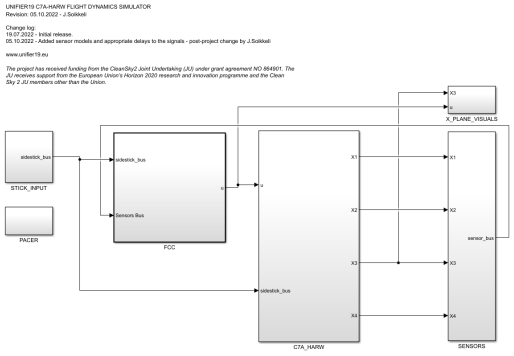


Fig. 11: UNIFIER19 Simulink Model

The SIMULINK model collaborates with MATLAB and Xplane 11 for visualization. These tools are our go-to for crafting and refining our control laws. Explore the details in the upcoming section.

B. FCC subsystem

The output of the FCC subsystem encompasses the command vector for surface control commands, auxiliary thrust commands, HTU command (main propeller), and, ultimately, the gear command. The principal objective of this subsystem is to furnish the command vector, and it achieves this goal through three distinct methodologies:

- Provide a constant command vector (initial command vector).
- Provide a vector calculated by our subsystems.
- Provide a vector from side-stick input.

Let us elucidate the figure below.

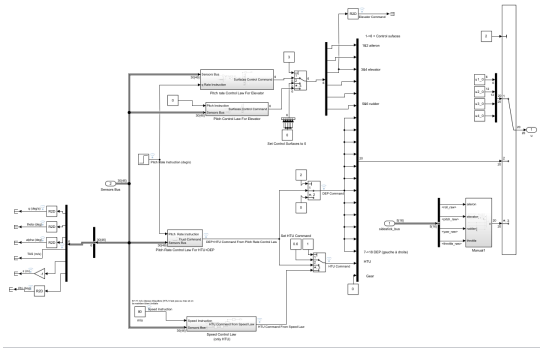


Fig. 12: FCC subsystem

Let's commence our explanation from the bottom-left corner, which serves as the focal point for reading sensor outputs. In this region, we can access data such as pitch rate (q), theta, alpha, True Airspeed (TAS), altitude (z), and phi. The ability to retrieve these values is facilitated through the demultiplexing of the sensor bus.
In the central upper section, the initial control law, titled "Pitch Rate Control Law for Elevator," is presented. Just below it lies the "Pitch Control Law for Elevator." Moving to the bottom center, we encounter the "Speed Control Law (only HTU)" subsystem, while above it resides the "Pitch Rate Control Law for HTU+DEP" subsystem. Detailed explanations of each of these subsystems will follow in subsequent sections, except for the last one, which remains incomplete.
Directly adjacent to the aforementioned area, three multiport switches are positioned, offering convenient toggling for activating or deactivating different laws to simulate various scenarios. These switches also enable the manual setting of direct commands. Subsequently, all these commands are consolidated through a multiplexer before reaching the final multiport switch.

Let's delve into the intricacies of our control laws.

C. Implementation of the pith-rate feedback loop

This subsystem takes two inputs: the pitch rate instruction and the sensor bus. The output is a single vector (1x6) containing commands for all control surfaces. The first two elements represent commands for the aileron, the next two for the elevator, and the last two for the rudder equivalent. Currently, our focus is solely on the elevator commands to facilitate pitch modifications.
Initially, we subtract the pitch rate instruction from the measured value (in radians per second); the resulting difference is logged. Subsequently, this difference is integrated to yield an angle. The resultant angle command undergoes correction via the PID block (further elaboration on this correction will follow shortly). Finally, the command passes through a saturation block to prevent excessive command angles, confined within the range of -30° to 30°.

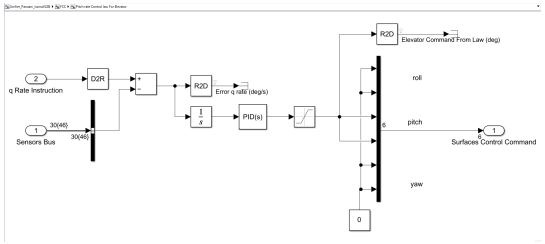


Fig. 13: Pitch-Rate Control Law

D. Reminder about Ziegler-Nichols tuning method for PID controller

All of our PID controllers are tuned using the Ziegler-Nichols method. This method is a relatively easy way to implement satisfying corrections with PID. Nevertheless, this is not the optimal method. We chose it regarding the difficulties to linearize the Unifier-19 model.
This method has three steps:

- First, only use proportional correction and find the gain which causes the oscillations of the system (Ker).
- Then measure the period of oscillations (Pcr).
- Finally, use the following table to tune your PID controller:

Controller	K_p	T_i	T_d
P	$0.5\ K_{cr}$	Infinity	0
PI	$0.45\ K_{cr}$	$P_{cr}/1.2$	0
PID	$0.6\ K_{cr}$	$P_{cr}/2$	$0.125\ P_{cr}$

Fig. 14: Ziegler-Nichols Method

E. Implementation of the pitch feedback loop

This feed back loop as the same input and output of the previous one. in fact the only differences is that there is no integrator after the subtraction.

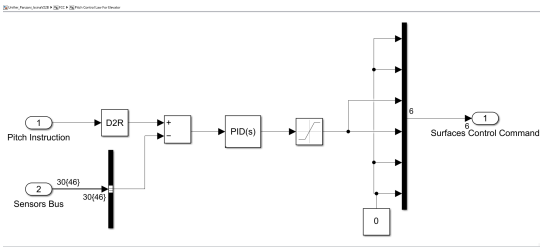


Fig. 15: Pitch Control Law

F. Implementation of the speed control law for HTU

The last subsystem developed is the speed control law. This block has two inputs; the first is the speed instruction, and the second is the sensors bus. The only output is a scalar between 0 and 1, which represents the thrust command of the main propeller, the HTU. As always, we begin by subtracting the instruction and the output of the TAS sensor. We correct our command with a PID controller, and finally, we use a saturation block to meet the constraint.

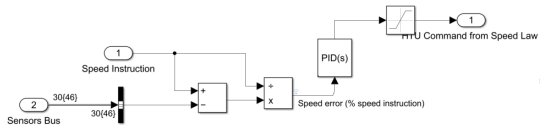


Fig. 16: Speed Control Law

To tune the PID controller of this loop, we activate the pitch control law and set the pitch angle instruction to zero (horizontal flight).

G. Last remarks on previous feedback loops

It's important to remember that our feedback loops are only valid under restricted conditions. For example, a pitch rate instruction of 1°/s will be reached, but not for a long time, depending on the TAS because of the stall. Additionally, we are aware that it is possible to design even more laws (on altitude, for example) or design in a different way (make the plane pitch up or down to reach speed instruction). In the end, it will be interesting to combine certain laws to achieve more complex ones.

V. Results & Discussion

In this section, we will present the response of our feedback control loops on pitch rate, pitch, and speed. Then we will present how we used it to show the effect of distributed propulsion.

A. Step response of control laws

For now, we haven't managed to linearize the Unifier-19 model, which is an important axis of improvement. With this linearization, we shall achieve better correction for all of our laws by using the controller tuning app of Simulink.
But, as said earlier, we used the Ziegler-Nichols method to tune our PID controller. Let's see the results, beginning with the main control feedback loop: the pitch rate.

1) Pitch rate: The two following figures show the same simulation; the simulation time is set to 60 seconds, the pitch rate instruction is 0.1°/sec, and it is important to notice that the instruction occurs at t=15s.

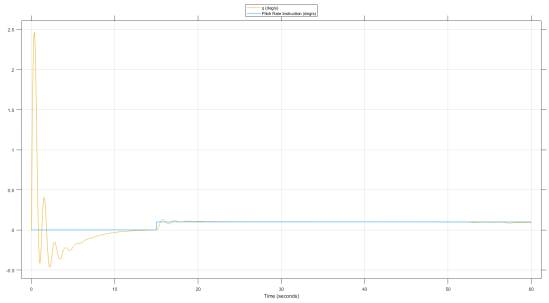


Fig. 17: Step response of Pitch Rate Control Law

In the first figure, we can see the whole simulation. The first observation that we can make is the significant oscillations that occur at the very beginning. The pitch rate reaches almost 2.5°/sec. We assume that these oscillations are due to the initial conditions of the model. We figure out that the magnitude of these oscillations is dependent on the initial command vector u0. We tried to modify it to see if any improvement was possible; however, the problem is that to find the best initial conditions that limit this magnitude, we need to use the Simulink trim app, which we were not able to run on this model yet. Any manual modifications were even worse. So, we decided for the rest of our study to wait for the end of this initial perturbation before setting our instruction. **This is why the instruction occurs at t=15s (valid for the rest of the study).**

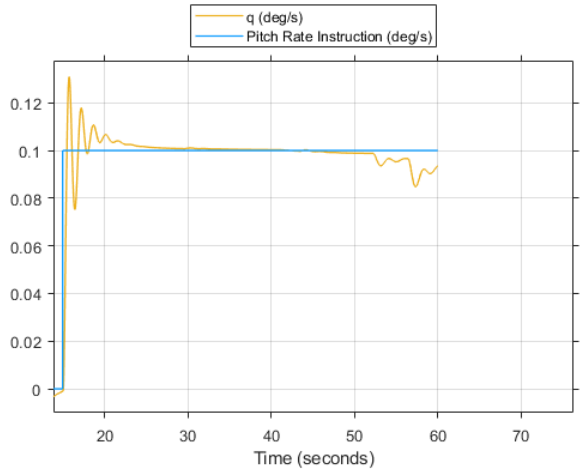


Fig. 18: Step response of Pitch Rate Control Law 2

The second figure focuses on the step response. As we can see, the maximum deviation is 0.025°/s, and the needed time to achieve a deviation of less than 10% is 4,92 seconds.
At the end of the simulation, we can see that the measured pitch rate starts to decrease, which is the beginning of the stall of the airplane. Indeed, with the initial perturbation, our theta angle is superior to zero since the beginning, then it continues to increase with the instruction; consequently, at t=50s, theta is superior to 0.1*(50-15)=3.5°. Plus, the simulation runs at TAS=67.71m/s, which is a low velocity. In these conditions, we can observe the start of the stall, which is predictable.
With the Xplane 11 visualization, we can conclude that the implementation of the pitch rate control law is successful.

2) Pitch: This feedback control loop has been design in order to design the last one which is the speed law. We will explain why after. The pitch control feedback is very close of the pitch rate. In fact the PID block as almost the same parameters.
The greatest deviation is 0.6°, and the time needed to achieve a deviation of less than 10% is 4.58 seconds.

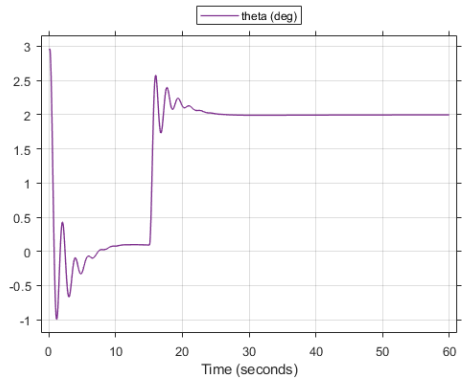


Fig. 19: Step response of Pitch Control Law

3) Speed: At the beginning, we tried to design this law without using the pitch control feedback. So when the instruction hits (TAS instruction = 80 m/s), the HTU thrust starts to increase. The pitch angle starts to increase too. Consequently, the speed evolution is badly affected, and Unifier-19 is not able to reach the speed instruction.
This is why we designed the previous law, and with it, we can reach the speed instruction.
The two next graphs represent the step response (TAS instruction = 80 m/s with initial TAS = 70 m/s) in velocity for a horizontal flight.
The maximum is less than 0.1 m/s, and the time constant is equal to 10 seconds.

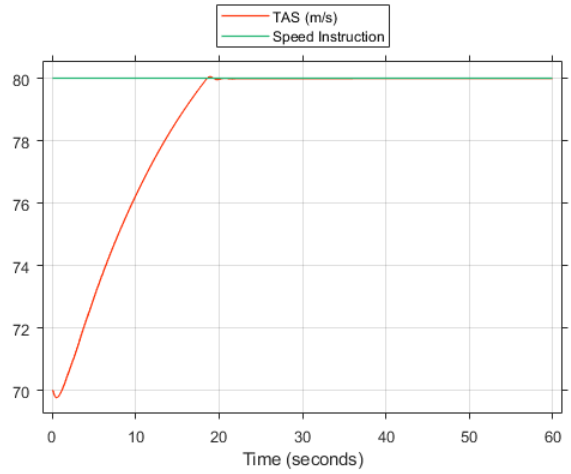


Fig. 20: Step response of Speed Control Law

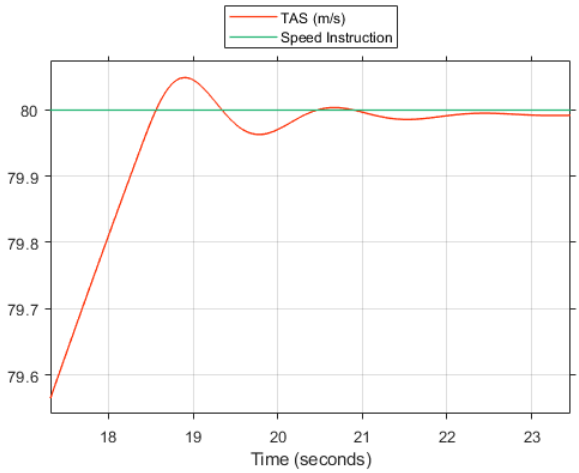


Fig. 21: Step response of Speed Control Law 2

After the design of these three laws, we wanted to use them to determine the effect of distributed propulsion.

B. *Effect of distributed propulsion on Unifier-19*

What we aim to demonstrate is the impact of distributed propulsion in the context of longitudinal movement, considering the aerodynamic configuration of Unifer-19. In order to do this, we concluded that the thrust commands of the twelve auxiliary propellers must be equal; an asymmetric distribution of thrust will perturb longitudinal stability. We use the pitch rate control law to show the effect.

1) Drag: Firstly we characterised the drag with and without DEP (distributed propulsion)

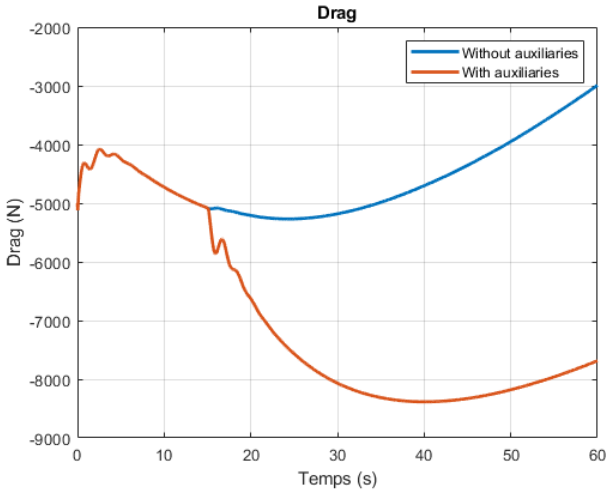


Fig. 22: Comparison of drag with and without DEP

It seems that the utilization of auxiliaries provokes a significant increase in drag $\Delta Fd = 3000N$ at t = 30s. But, let's take a moment to check the velocity profile.

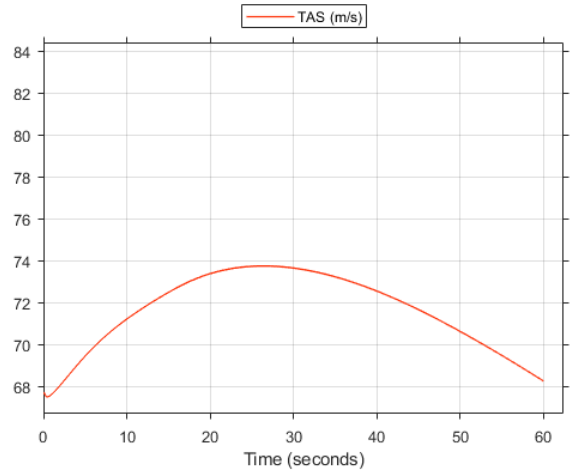


Fig. 23: Velocity profile without DEP

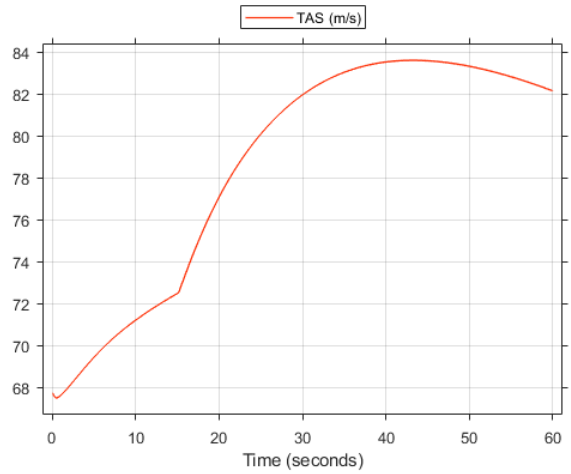


Fig. 24: Velocity profile with DEP

We'll remind you of the drag equation:

$$F_d = \frac{1}{2}\rho v^2 C_d A$$

The difference in velocity should explain a large part of the drag increase with DEP.
That's why we designed the TAS control feedback loop. Let's use it and compare the velocity profile in both situations.

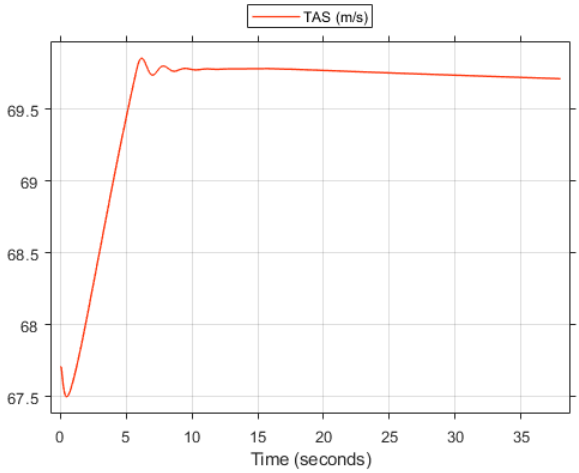


Fig. 25: Velocity profile without DEP with TAS control loop active

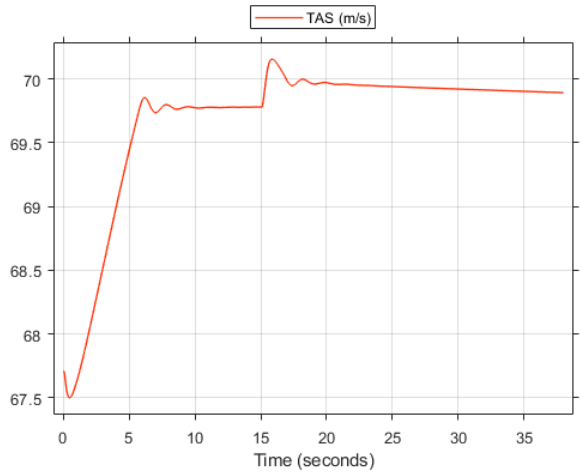


Fig. 26: Velocity profile with DEP with TAS control loop active

With the speed instruction set to 70 m/s, both profile are close to each other, especially between $25 < t < 30s$. Finally let's evaluate the drag with this new condition.

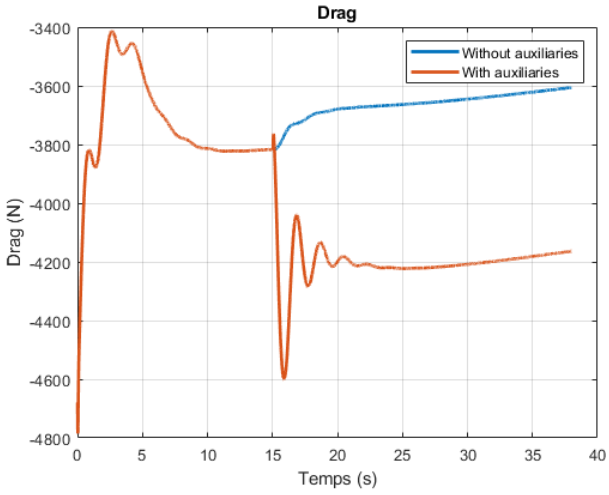


Fig. 27: Comparison of drag with and without DEP at constant speed

As predicted the TAS control law allowed the reduction of the drag difference. Nevertheless we still observe that DEP induce extra drag $\Delta Fd = 600N$ at t = 30 s. Let's see the elevator angle in both situation.

2) Elevator command angle: Here, we want to see if the additional drag with DEP is at least partially attributable to the elevator angle. The following graph shows the elevator command angle (bottom), the pitch rate (left), and the DEP thrust command (in this case 0).

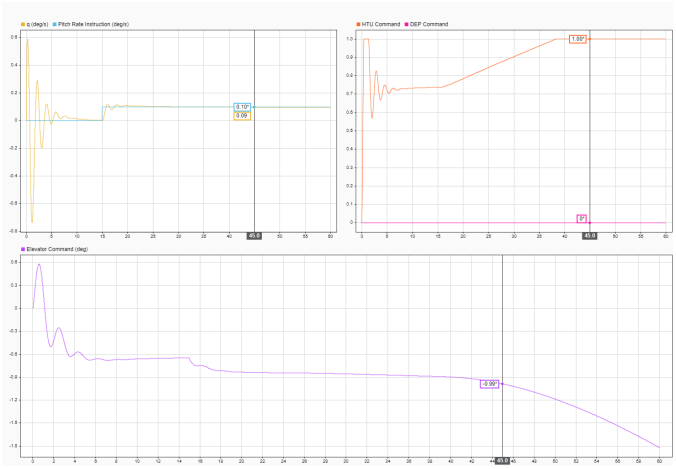


Fig. 28: Elevator command angle without DEP

Then, here is the same simulation, but DEP thrust is no longer equal to zero (DEP command = 1).

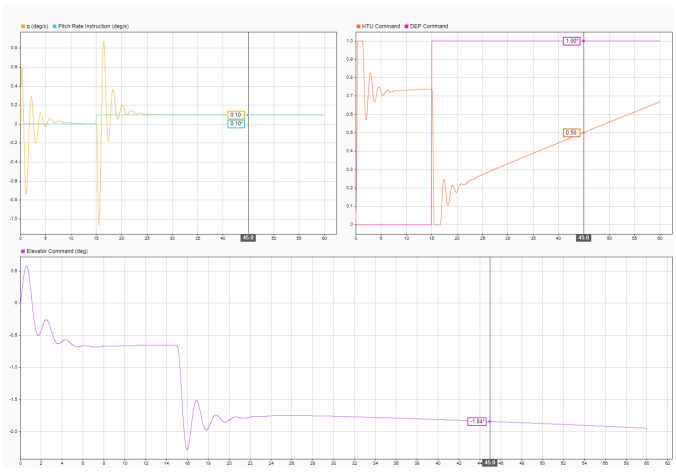


Fig. 29: Elevator command angle with DEP

It seems that the use of DEP increases the elevator angle (by one degree in these simulations). This larger angle induces more drag, which partially explains the additional drag seen in the previous section.
Nevertheless, we can also observe that DEP can be useful here. Indeed, as a source of additional thrust, it delays the moment when UNIFIER-19 cannot maintain its velocity and continues to pitch up. Consequently, the elevator angle starts increasing rapidly; that's what we observe at the end of simulation one. On the other hand, simulation two shows an approximately constant angle until the end.

VI. Conclusion

This project allowed us to familiarize ourselves with new tools such as Simulink Aerospace Blockset and other special features of MATLAB. It also gave us the opportunity to delve deeper into flight mechanics, automation, and, thanks to it, we discovered an avant-garde technology, which is distributed propulsion.

Our work doesn't prove that distributed propulsion for UNIFIER-19 in the case of longitudinal movement is not interesting but our study on its drag does not show any clear improvement and rather indicates a sign of degradation, although it would, of course, be necessary to conduct more simulations with a more reliable model.

It will be very interesting to continue Jeanne Trias's[4] work on lateral movement using distributed propulsion because this type of propulsion has the potential to bring significant benefits in this context.

Acknowledgment

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